

Problem Set 4: Questions on Random Sampling

Statistics 1: Data Exploration

1. Define simple random sampling without replacement (SRSWOR) and simple random sampling with replacement (SRSWR). Explain how they differ. Give one real example where SRSWOR is preferable.
2. State and prove that under SRSWOR, each sample of size n from a population of size N has equal probability of being selected.
3. Let $Y_1, Y_2, \dots, Y_n$ be an SRSWOR from a population with values $y_1, \dots, y_N$. Show that the sample mean $\bar{Y}$ is an unbiased estimator of the population mean $\bar{Y}_N$.
4. Under SRSWOR, write the variance of the sample mean:

$$\text{Var}(\bar{Y}) = \left(1 - \frac{n}{N}\right) \frac{S^2}{n},$$

where S^2 is the population variance. Explain the role of the finite population correction (FPC) factor and when it becomes negligible.

5. Differentiate between sampling error and non-sampling error. Provide two examples of non-sampling errors commonly encountered in large national surveys such as NFHS or NSS.
6. A population consists of the values:

$$\{5, 8, 12, 15\}.$$

List all possible SRSWOR samples of size $n = 2$. Compute:

- (a) the mean of each possible sample,
- (b) the average of these sample means,

and verify that it equals the population mean.

7. A population contains the values:

$$\{10, 20, 30, 40, 50\}.$$

An SRSWOR sample of size $n = 3$ yields the observed sample:

$$\{20, 30, 50\}.$$

Compute:

- (a) the sample mean,

- (b) the sample variance (with denominator $n - 1$),
- (c) the unbiased estimator of the population variance under SRSWOR.

8. Consider a population of size $N = 6$ with values:

$$\{2, 4, 6, 8, 10, 12\}.$$

If an SRSWR sample of size $n = 3$ is drawn, what is the probability that all three sampled values are equal to 12?

9. A population of size $N = 5$ has values:

$$\{3, 6, 9, 12, 15\}.$$

Under SRSWOR, a sample of size $n = 2$ yields:

$$\{6, 12\}.$$

Compute:

- (a) an unbiased estimate of the population total,
- (b) the variance of this estimator under SRSWOR.

10. A population of size $N = 7$ has values:

$$\{4, 7, 10, 13, 16, 19, 22\}.$$

An SRSWOR sample of size $n = 3$ gives observed values:

$$\{7, 13, 19\}.$$

Compute:

- (a) the sample mean,
- (b) the estimate of the population mean,
- (c) the standard error of the estimator of the mean using the finite population correction (FPC).